Interop QOS

2026-03-24-13-07-14

Objective

The objective of the QoS (Quality of Service) traffic throughput test is to measure the maximum achievable throughput of a network under specific QoS settings and conditions. By conducting this test, we aim to assess the capacity of network to handle high volumes of traffic while maintaining acceptable performance levels, ensuring that the network meets the required QoS standards and can adequately support the expected user demands.

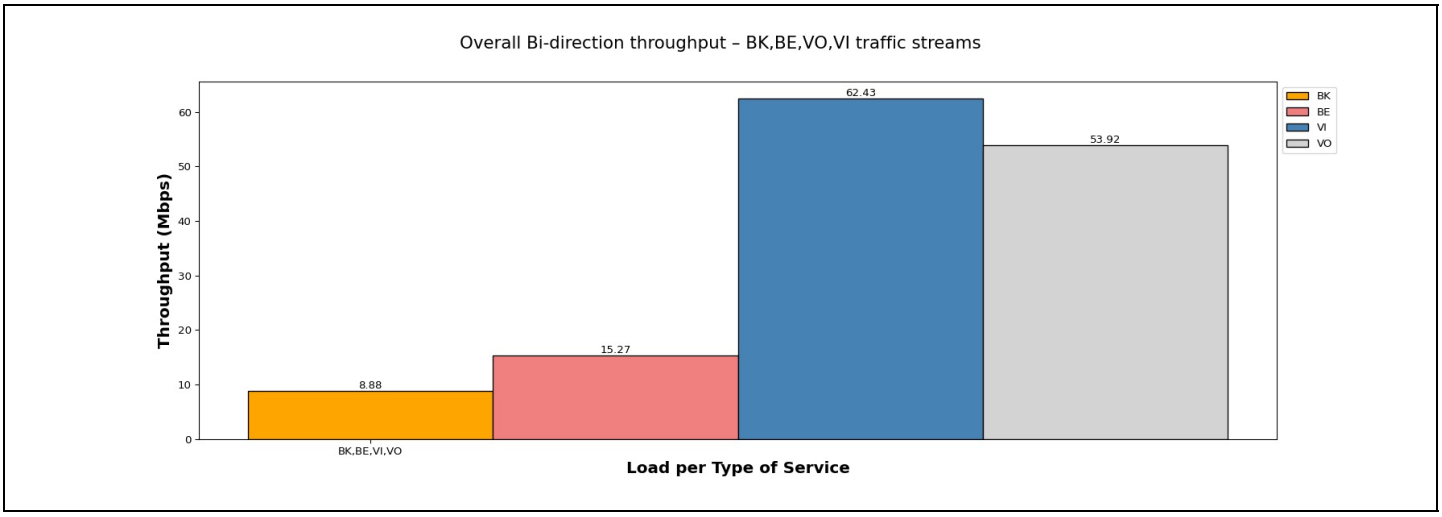
Test Configuration	Device List	vivonew(Android), ubuntu24(Linux), macoss-MacBook-Air.local(Mac), sungsung(Android)
	Number of Stations	Total(4) Android(2) Linux(1) Mac(1)
	AP Model	NETGEAR
	SSID	None
	Traffic Duration in hours	0.02
	Security	None
	Protocol	TCP
	Traffic Direction	Bi-direction
	TOS	['BK', 'BE', 'VI', 'VO']
	Per TOS Load in Mbps	Upload:100.0 Mbps,Download:100.0 Mbps

Overall Bi-direction Throughput for all TOS i.e BK | BE | Video (VI) | Voice (VO)

No of Stations	Throughput for Load 100.0 Mbps-upload	Throughput for Load 100.0 Mbps-download
4	BK : 5.0, BE : 7.0, VI: 30.53, VO: 30.92	BK : 3.88, BE : 8.27, VI: 31.9, VO: 23.0

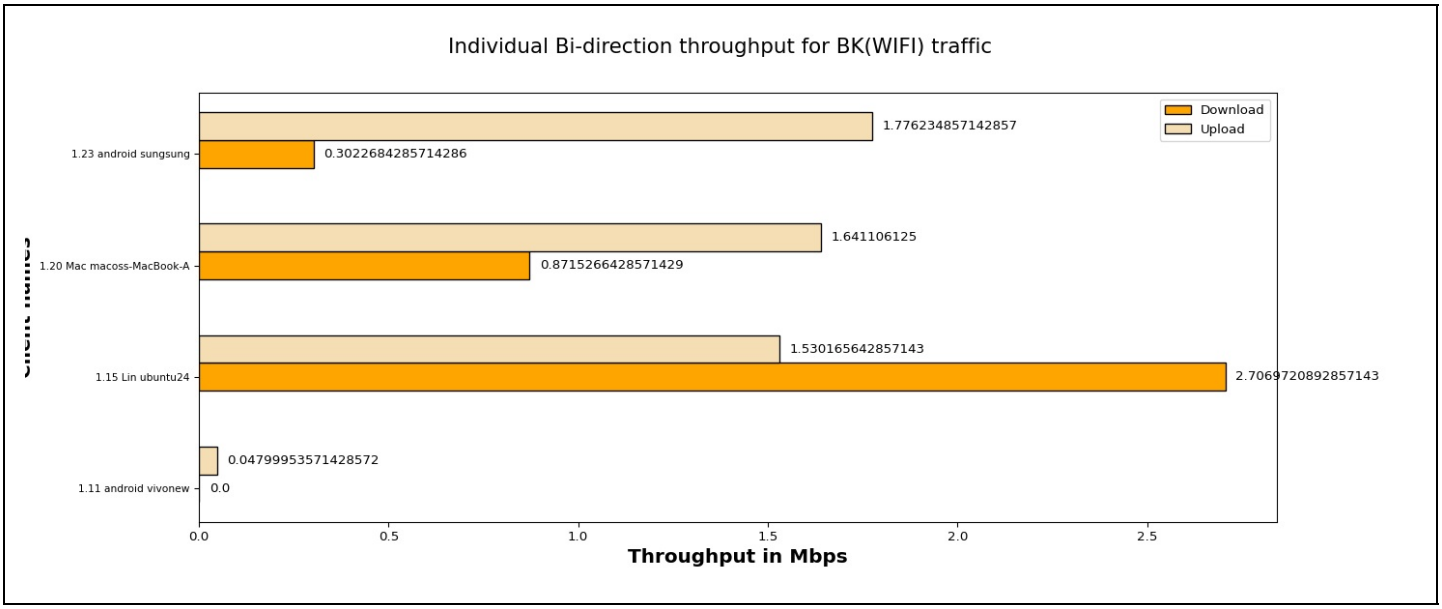
Overall Bi-direction throughput for 4 clients with different TOS.

The below graph represents overall Bi-direction throughput for all connected stations running BK, BE, VO, VI traffic with different intended loadsUpload:100.0 Mbps,Download:100.0 Mbps per tos



Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic BK(WiFi).

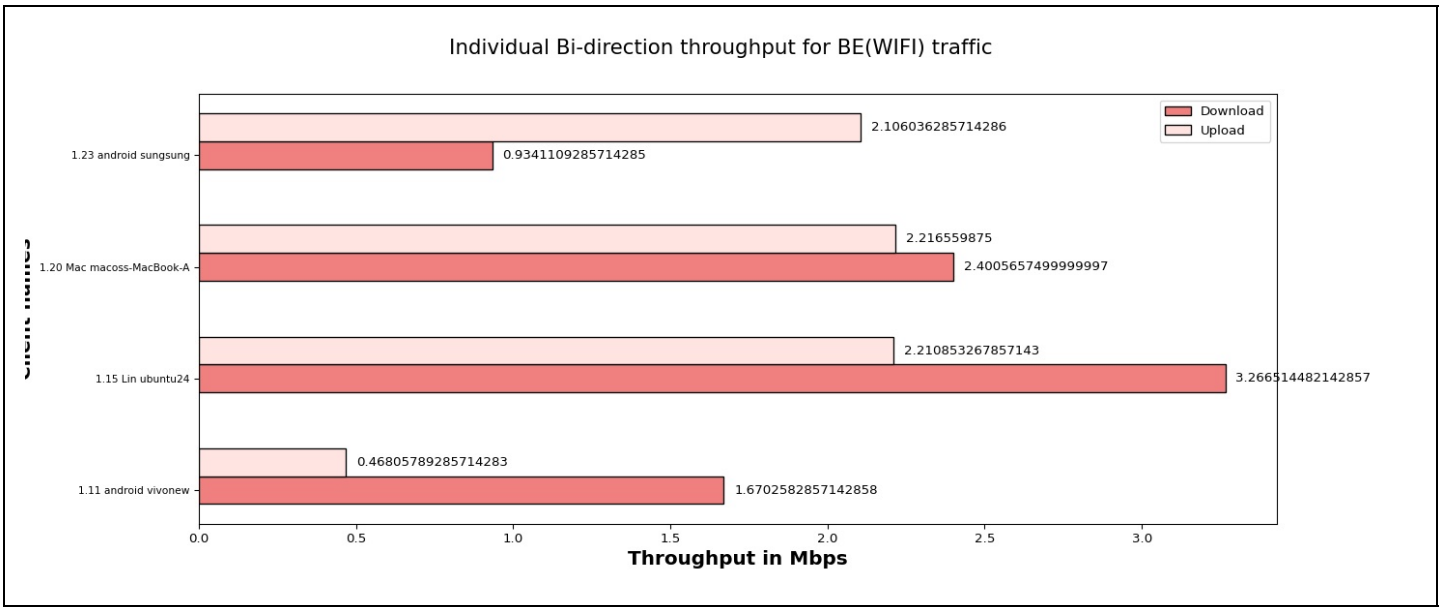
The below graph represents individual throughput for 4 clients running BK (WiFi) traffic. X- axis shows “Throughput in Mbps” and Y-axis shows “number of clients”.



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.11 android vivonew	92:da:84:1a:a3:75	<unknown ssid>	Background	100.0 Mbps	100.0 Mbps	0.04799953571428572 Mbps	0.0 Mbps	13.281250	14.772727
1.15 Lin ubuntu24	c:5f:67:a1:39:86	NETGEAR_5G_wpa2	Background	100.0 Mbps	100.0 Mbps	1.530165642857143 Mbps	2.7069720892857143 Mbps	10.678426	12.209023
1.20 Mac macoss-MacBook-Air.local	30:35:ad:c5:e1:be	NETGEAR_5G_wpa2	Background	100.0 Mbps	100.0 Mbps	1.641106125 Mbps	0.8715266428571429 Mbps	1.452244	31.838134
1.23 android sungsung	ca:84:bc:fa:cb:7a	NETGEAR_5G_wpa2	Background	100.0 Mbps	100.0 Mbps	1.776234857142857 Mbps	0.3022684285714286 Mbps	25.037416	29.101295

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic BE(WiFi).

The below graph represents individual throughput for 4 clients running BE (WiFi) traffic. X- axis shows “number of clients” and Y-axis shows “Throughput in Mbps”.

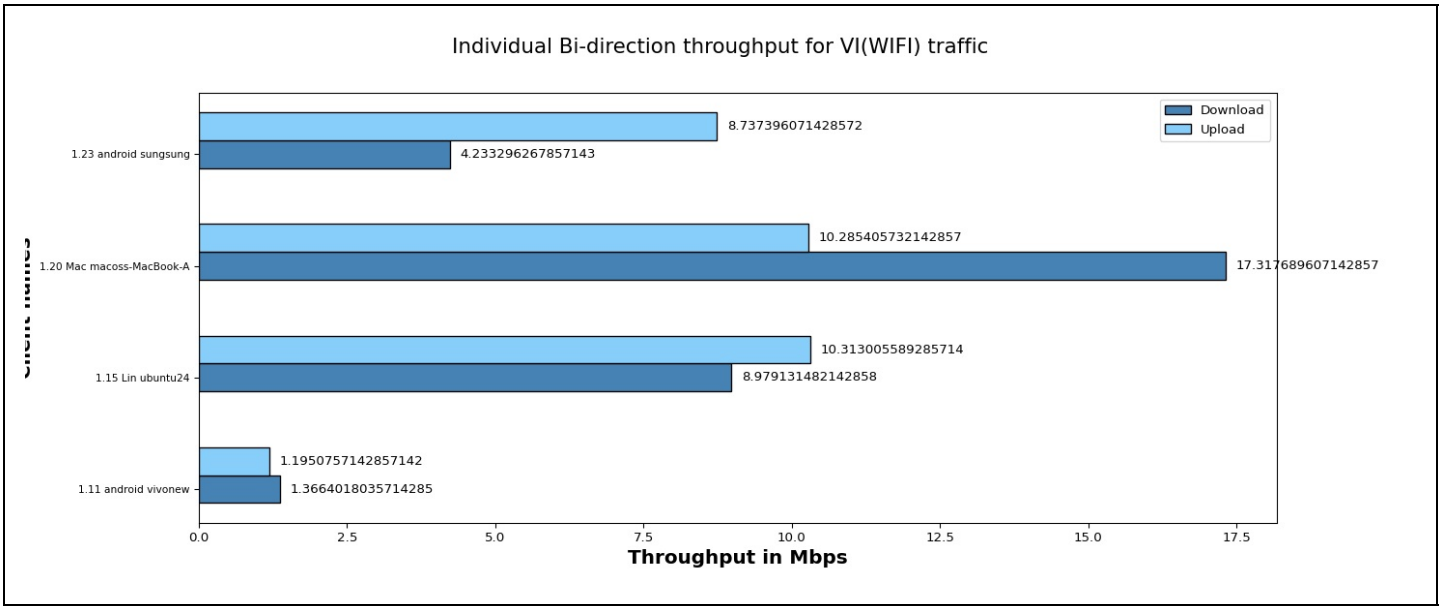


Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.11 android vivonew	92:da:84:1a:a3:75	<unknown ssid>	Besteffort	100.0 Mbps	100.0 Mbps	0.46805789285714283 Mbps	1.6702582857142858 Mbps	14.882361	18.404615
1.15 Lin				100.0	100.0	2.210853267857143	3.266514482142857		

ubuntu24	c:5f:67:a1:39:86	NETGEAR_5G_wpa2	Besteffort	Mbps	Mbps	Mbps	Mbps	4.548505	4.447476
1.20 Mac macoss-MacBook-Air.local	30:35:ad:c5:e1:be	NETGEAR_5G_wpa2	Besteffort	100.0 Mbps	100.0 Mbps	2.216559875 Mbps	2.4005657499999997 Mbps	0.515842	6.629554
1.23 android sungsung	ca:84:bc:fa:cb:7a	NETGEAR_5G_wpa2	Besteffort	100.0 Mbps	100.0 Mbps	2.106036285714286 Mbps	0.9341109285714285 Mbps	9.151343	25.678255

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic VI(WiFi).

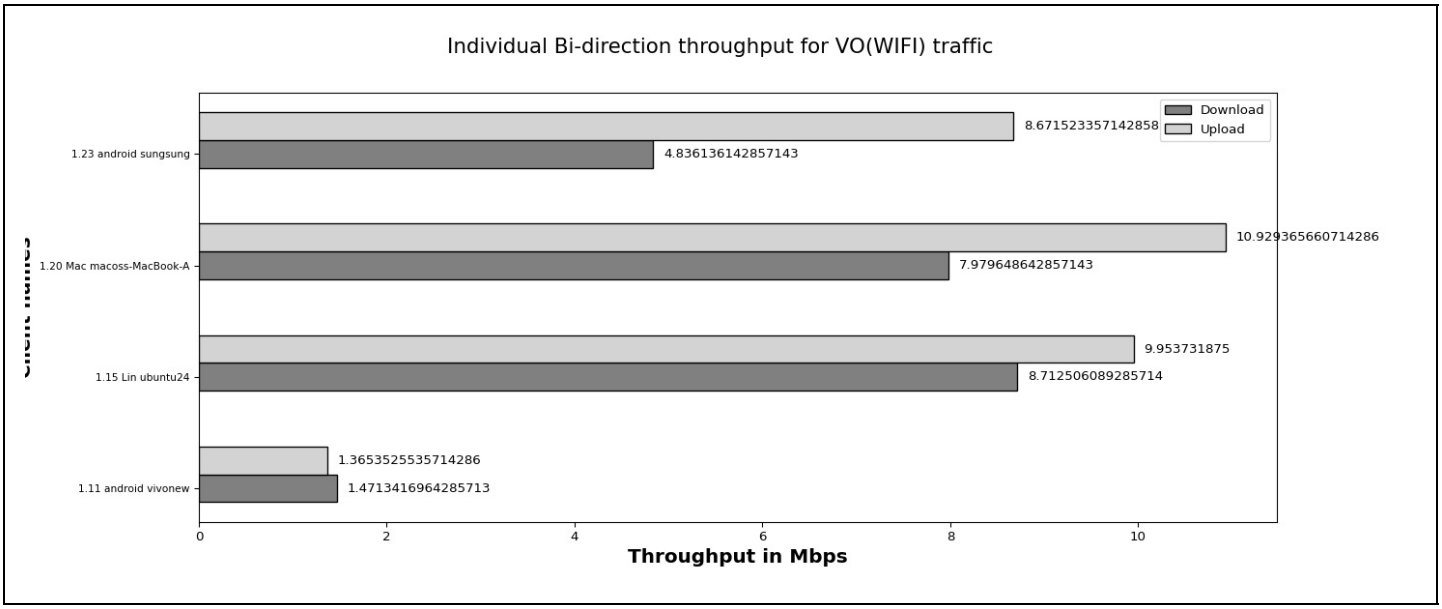
The below graph represents individual throughput for 4 clients running VI (WiFi) traffic. X- axis shows "number of clients" and Y-axis shows "Throughput in Mbps".



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.11 android vivonew	92:da:84:1a:a3:75	<unknown ssid>	Video	100.0 Mbps	100.0 Mbps	1.1950757142857142 Mbps	1.3664018035714285 Mbps	10.586690	14.509029
1.15 Lin ubuntu24	c:5f:67:a1:39:86	NETGEAR_5G_wpa2	Video	100.0 Mbps	100.0 Mbps	10.313005589285714 Mbps	8.979131482142858 Mbps	2.724062	2.584402
1.20 Mac macoss-MacBook-Air.local	30:35:ad:c5:e1:be	NETGEAR_5G_wpa2	Video	100.0 Mbps	100.0 Mbps	10.285405732142857 Mbps	17.317689607142857 Mbps	0.256741	0.568147
1.23 android sungsung	ca:84:bc:fa:cb:7a	NETGEAR_5G_wpa2	Video	100.0 Mbps	100.0 Mbps	8.737396071428572 Mbps	4.233296267857143 Mbps	1.292450	11.860500

Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic VO(WiFi).

The below graph represents individual throughput for 4 clients running VO (WiFi) traffic. X- axis shows "number of clients" and Y-axis shows "Throughput in Mbps".



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.11 android vivonew	92:da:84:1a:a3:75	<unknown ssid>	Voice	100.0 Mbps	100.0 Mbps	1.3653525535714286 Mbps	1.4713416964285713 Mbps	7.610046	14.497905
1.15 Lin ubuntu24	c:5f:67:a1:39:86	NETGEAR_5G_wpa2	Voice	100.0 Mbps	100.0 Mbps	9.953731875 Mbps	8.712506089285714 Mbps	1.169522	1.599313
1.20 Mac macoss-MacBook-Air.local	30:35:ad:c5:e1:be	NETGEAR_5G_wpa2	Voice	100.0 Mbps	100.0 Mbps	10.929365660714286 Mbps	7.979648642857143 Mbps	0.175256	3.717950
1.23 android sungsung	ca:84:bc:fa:cb:7a	NETGEAR_5G_wpa2	Voice	100.0 Mbps	100.0 Mbps	8.671523357142858 Mbps	4.836136142857143 Mbps	0.757025	6.704907

Information	contact	support@candelatech.com